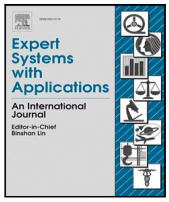
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Target hierarchy-guided knowledge tracing : Fine-grained knowledge state modeling

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ABSTRACT

Knowledge Tracing (KT) focuses on modeling the exercise process of students, assessing their knowledge state changes during the exercise process, and further providing targeted guidance for teaching and learning. Current KT models have made significant progress by utilizing deep learning technologies to model relevant attributes of the questions. However, in current knowledge tracing strategies, we have yet to exhaustively explore how to precisely evaluate the hierarchies of students' knowledge mastery to better discern their current target hierarchy. To address the limitations of the KT model in accurately identifying and positioning target hierarchy, our research focuses on two key areas: first, we dynamically track students' target hierarchies by evaluating their feedback on questions at different hierarchies ; second, we assess their actual question-solving abilities within these target hierarchies, thereby optimizing our evaluation of their knowledge status. Based on these insights, we propose the ***T**arget **H**ierarchy-guided **K**nowledge **T**racing* (***THKT***) model. In this model, we first incorporate the identification of question hierarchy into the model representation. Then, to avoid homogenization of the target hierarchy, we dynamically track the appropriate hierarchy for students based on their varying feedback to questions of the same hierarchy, thereby pinpointing their learning target hierarchies. Simultaneously, we develop a KCs applied ability learning module that works in conjunction with the target hierarchies to generate interpretable prediction results. The proposed THKT model has been tested and evaluated using three public, real-world educational datasets. The findings clearly demonstrate that our approach shines in the realm of KT prediction tasks, providing significantly interpretability. For broader research, we plan to provide the source code at: https://github.com/xinjiesun-ustc/THKT to ensure accessibility and promote further innovation in this field.

1. Introduction

Knowledge tracing (KT) is a method that simulates students' mastery of knowledge states (e.g., irrational numbers, rational numbers) over time, based on their historical practice (e.g., responses to a series of exercises), thereby predicting future performance (Ghosh, Heffernan, & Lan, 2020; Liu, Shen, Huang, Chen, & Zheng, 2021; Rejeb, Abdollahi, Rejeb, & Mostafa, 2023). Fig. 1 provides a toy example of knowledge tracing, where the student's knowledge state continuously changes as the exercise sequence deepens. This change can effectively feed back into teaching and enhance learning outcomes (Harrell, 2013). KT is a classic task in the field of intelligent education and is widely used in online teaching platforms (Mongkhonvanit, Kanopka, & Lang, 2019), interactive learning (Liu, Tong, et al., 2019), and intelligent tutoring systems (Huang et al., 2017). Therefore, it has immense academic and practical value.

Student knowledge states are typically dynamically evaluated by synthesizing some direct or indirect indicators generated in the student's exercise sequence. For example, early researchers (Choi et al., 2020; Minn, Yu, Desmarais, Zhu, & Vie, 2018) only used responses to Knowledge Components (KCs) to model changes in student knowledge states, which often led to inaccurate estimates of student knowledge states. Researchers like Liu, Huang, et al. (2019), Pandey and Srivastava (2020) and Tong et al. (2020) proposed using both problem and KCs information to construct the evolution of student knowledge states during the exercise process. To better simulate and understand the student learning process, memory structures (Abdelrahman & Wang,

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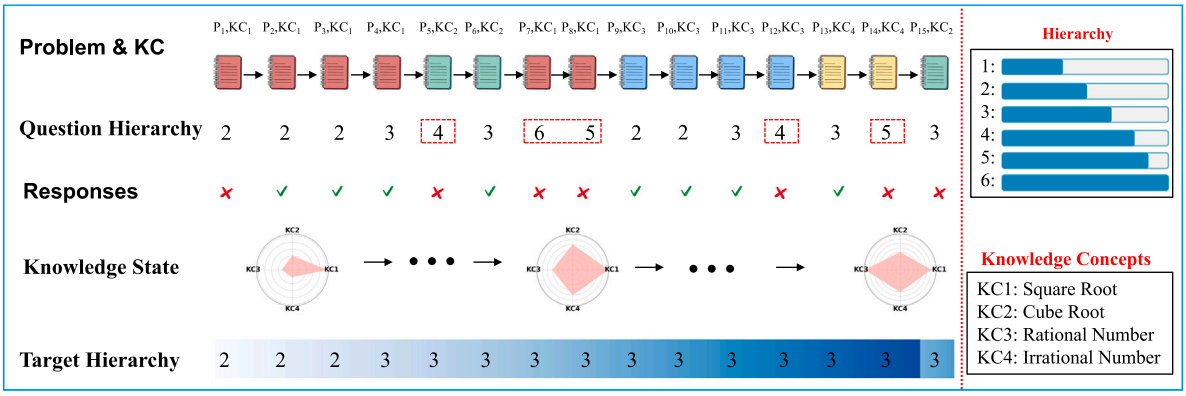


Fig. 1. A toy example of students' exercise process and knowledge tracing. The value of hierarchy refers to the hierarchy to which the problem and knowledge concept belongs. The topics marked by the red dashed rectangular box are of a higher hierarchy. In this figure, *Question Hierarchy* represents the hierarchy of the question attempted by students, which is derived from training that incorporates students' problem IDs, KCs, and KCH (hierarchy of KCs based on prior statistics). Within *Target Hierarchy*, a darker color indicates a higher probability of assigning the student to that particular hierarchy at a given moment. The length of the hierarchy bar symbolizes the strength of the hierarchy; the longer the bar, the stronger the hierarchy, conversely, the shorter the bar, the weaker the hierarchy.

2019; Zhang, Shi, King, & Yeung, 2017), forgetting perceptions (Shen, Chen, et al., 2022; Shen et al., 2021), etc., take the time dimension into account to track changes in student knowledge states. However, these methods only go as far as modeling students' knowledge performance when assessing their knowledge state, failing to further delve into the issue of knowledge hierarchy.

As depicted in Fig. 1, during the initial stage of the exercise sequence, the student shows a solid grasp on KC of 1 (square root). However, a series of errors subsequently emerges, this phenomenon is intuitively perplexing. Recent studies have attempted to address this issue by calculating question difficulty (Liu et al., 2020; Shen, Huang, et al., 2022; Zhang et al., 2021) or employing dynamic classification methods (Minn, Desmarais, Zhu, Xiao, & Wang, 2019; Minn et al., 2018). Although these approaches can somewhat explain the inconsistent performance of students on the same KC, they also exhibit certain limitations. The calculation of question difficulty is tailored to each specific question, yet the perceived difficulty of the same question may vary among students. Dynamic classification methods may cause students' knowledge states to be excessively influenced by their group peers, leading to a lack of personalization. It is particularly noteworthy that Koedinger, Pavlik, Stamper, Nixon, and Ritter (2010) and Stamper, Koedinger, and Mclaughlin (2013) argue that a knowledge component might actually encompass multiple sub-components. Therefore, the inconsistency in a student's performance on a given knowledge component could be due to varying levels of mastery over these sub-components. This perspective should be widely recognized in the field of knowledge tracing, as many problems indeed involve multiple knowledge points, yet the existing solutions often only consider the first knowledge point involved in the problem.

In the realm of education, it is crucial to understand and track the hierarchical nature of students' knowledge mastery, such as Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating, as outlined by Momen, Ebrahimi, and Hassan (2022), and to conduct personalized positioning based on this hierarchical understanding. For instance, revisiting Fig. 1, the student's answers to questions exhibit weak intuitive interpretability, with alternating correct and incorrect responses on the same KC and elusive relationships between different KCs. However, if considering the hierarchy of each question, we can observe that when the question hierarchy reaches 4 or higher, even if the traditional KT model predicts a satisfactory mastery of the KC, the student often fails to provide valid answers. The target hierarchy learning mechanism can continuously track his/her feedback on different hierarchies of question in the learning record, and dynamically update his/her target hierarchy. From this, we can infer that the student is currently capable of solving questions at third hierarchy or below. Consequently, the student's exercise target hierarchy should be temporarily set at the third hierarchy, meaning that questions at or below the third hierarchy should be their primary focus. Particularly in complex subject areas such as mathematics and science, knowledge mastery often involves multiple hierarchies. In such cases, traditional KT methods may neglect the difficulties students may encounter while understanding and mastering various question hierarchies. Therefore, a novel approach is required to deepen our understanding of personalized target hierarchy knowledge tracking for individual students.

Considering the different learning history of different students, it is necessary to accurately identify the target hierarchy and the differences in specific domain knowledge for each individual student in order to precisely evaluate their current status. However, to properly attribute the target hierarchy and use it reasonably, three key challenges need to be addressed. ***Firstly***, how to ensure that the generated question hierarchy has a high degree of credibility. Given the long-tail problem that has always existed in educational datasets, hierarchical judgments based on problems in the student's exercise sequence will be limited. ***Secondly***, how to accurately divide the personalized target hierarchy based on the generated question hierarchy and the unique learning history of different students. That is, the question hierarchy allocation based on prior statistics is not always suitable for all students' situations. ***Lastly***, how to better use the target hierarchy at which the student is located to grasp the ability of the student to solve specific problems. To address these issues, a fine-grained operation method is needed that takes into account both the personalized student target hierarchy and the student's ability to apply knowledge. This will ensure a more target judgment of the student's knowledge state.

To address the challenges outlined above, we propose an ***Target Hierarchy-guided Knowledge Tracing*** (***THKT***) model. This KT method utilizes target hierarchies to guide the assessment of students' knowledge states. The model introduces several key innovations to enhance the effectiveness, adaptability, and interpretability of KT. **Firstly**, in order to enhance the credibility of question hierarchies, we start by learning the hierarchies of KCs (KCH) based on the accuracy and frequency of KCs in prior statistics. **Secondly**, to compensate for the rigidity of KCH conducted through prior statistics, we further introduce a question hierarchy identification network to calculate the question hierarchy with fine-grained details by combining different problem information. Following this, we adopt a dynamic tracking attention mechanism for target hierarchies, to ensure that future question predictions pay more attention to the student's personalized learning history, thereby achieving a more accurate determination of target hierarchies. **Lastly**, we combine the target hierarchy where the students are located with the actual application skills they demonstrate in the question

practice sequence, to comprehensively judge the students' knowledge state at the current target hierarchy.

In summary, our proposed THKT model contributes in multiple ways: it introduces the learning history to make pre-judgments about the target hierarchies, combines this with students' question-solving abilities, addresses key challenges in KT task, and effectively explains students' future performance. The main contributions of this study can be summarized as follows:

- To the best of our knowledge, this is the first time students have been assigned personalized target hierarchies. We stratify students' target hierarchies in three stages: building upon the initial knowledge concepts hierarchy (KCH) that is derived from prior statistics, and the question hierarchy (QH) which is based on feedback from historical learning data, a dynamic tracking process is implemented for the target hierarchy (TH).
- We have carefully designed the THKT model to consider both the target hierarchies of students and their ability to solve practical questions during the learning process. This combination allows us to construct accurate, personalized representations of students' knowledge states.
- Extensive experimental results demonstrate the effectiveness of the THKT model and the good adaptability of the proposed target hierarchies. Furthermore, the THKT exhibits excellent interpretability, as it can clearly explain the reasons behind students' responses to problems.

2. Related works

Accurate identification of students' learning targets, pinpointing their respective target hierarchies, and effectively merging these hierarchies with students' knowledge application capabilities is a meaningful endeavor. Subsequently, we will delve into related work from two perspectives: educational target hierarchy and knowledge tracing.

2.1. Educational target hierarchy

Over the past few decades, the hierarchy and definition of educational targets have become important topics in educational research (Dlamini & Gerber, 2023; Kaviyaraj & Uma, 2022; Kemp, Palmer, & Strelan, 2019). This is because clearly and precisely defining educational targets can help teachers, students, and educational policy makers better understand and implement the educational process.

The early system of educational target hierarchy was pioneered by Bloom's Taxonomy (Forehand et al., 2005), proposed by Bloom and others in 1956. This taxonomy categorizes educational targets into three major types: cognitive targets, affective targets, and psychomotor targets, each subdivided into several hierarchies. This classification has had a profound impact on subsequent educational research. However, early educational target classifications were relatively simple. To address these issues, recent years have seen some evolutions of Bloom's taxonomy. For instance, Barari, RezaeiZadeh, Khorasani, and Alami (2022) and Bloom, Engelhart, Furst, Hill, and Krathwohl (1956) proposed a revised version of Bloom's taxonomy, which restructures the original hierarchy to better reflect new educational demands. In the field of education, it is crucial to effectively analyze and stratify educational targets. This can help educators and learners understand learning targets more clearly, thus enabling more effective design and implementation of teaching strategies. Many researchers have proposed a variety of methods to achieve this goal. For example, the theory of Kruglanski primarily focuses on how people "seize" and "freeze" when confronted with information, that is, how they make choices and decisions when faced with a large amount of information (Kruglanski & Webster, 1996). Maslow's hierarchy of needs is a widely known psychological theory, which proposes that human needs can be divided into five hierarchies (Newman & Newman, 2022). Gardner's theory of multiple intelligences emphasizes that education should value and develop a variety of intelligences in students (Hasanuddin & Siregar, 2022).

The application of educational target hierarchy is extremely broad. For example, Pikhart and Klimova (2019) applied target hierarchy in the application of linguistics in blended learning. Adijaya, Widiana, Agung Parwata, and Suwela Antara (2023) used Bloom's taxonomy to enhance programming capabilities and learning outcomes. Ahmed, Lashari, and Golo (2023) evaluated national single-course primary English textbooks from the perspective of Bloom's taxonomy. Although these studies adeptly establish various educational target hierarchies to accommodate the evolution of teaching, determining the precise target hierarchy a student falls within necessitates further exploration. In our current research, we employ deep learning methodologies to dynamically track and assess the specific target hierarchy of each student.

In summary, the method of educational target hierarchy plays a crucial role across various educational fields (Momen et al., 2022). Appropriate hierarchy can help teachers organize and impart knowledge more effectively, enabling students to understand and absorb information better. Our study classifies educational objectives into six hierarchies, inspired by Bloom's Taxonomy of Educational Objectives. However, the question hierarchies identified by the model may not fully correspond to Bloom's categorization of questions(remembering, understanding, applying, analyzing, evaluating, and creating). The higher the hierarchy, the greater the question-solving ability it represents. Furthermore, we establish the relationship between a student's application abilities and target hierarchies to better assess the student's dynamic knowledge state.

2.2. Knowledge tracing

With the burgeoning growth of online learning, research in the field of Knowledge Tracing (KT) has seen extensive exploration and development (Abdelrahman, Wang, & Nunes, 2023; Huang et al., 2024; Sun et al., 2022). The pioneering model in KT was the Bayesian Knowledge Tracing (BKT), which employs a hidden Markov model to characterize a student's knowledge state (Corbett & Anderson, 1994). Moreover, some factor analysis models, such as Factorization Machines (FMs) (Vie & Kashima, 2019), incorporate existing educational literature models as special cases. These notably include the additive factor model, performance factor model, and multidimensional item response theory.

In recent years, with the rapid development of neural networks, they have been utilized to track the evolution of students' knowledge states. Specifically, Deep Knowledge Tracing (DKT) (Piech et al., 2015) was the first to introduce the deep learning model of Recurrent Neural Networks (RNNs) into KT, achieving promising results. Subsequently, a plethora of models based on RNNs emerged, such as those modified by Lee and Yeung (2019), Liu, Huang, et al. (2019) and Yu et al. (2019) who altered the basic structure of the model. Others, like Cheung and Yang (2017), Sonkar, Waters, Lan, Grimaldi, and Baraniuk (2020) and Wang, Feng, Tang, Huang, and Liu (2019), added additional question-solving features, while (Minn et al., 2018; Shen, Chen, et al., 2022) simulated educational theories, among others. Then, models like DKVMN (Zhang et al., 2017), DKVMN-DT (Sun et al., 2019), and DKVMN-LA (Sun et al., 2021) used memory networks to store and continuously update students' knowledge states. These sequence modeling methods have achieved great success, but they often struggle to capture long-distance knowledge states.

The Self-Attentive Knowledge Tracing (SAKT) model (Pandey & Karypis, 2019) is capable of modeling each step in a student's learning process, considering not only the student's past correct or incorrect answers but also the order of these responses, while taking into account the interrelationships between these steps. The Attentive Knowledge Tracing (AKT) model (Ghosh et al., 2020) employs a novel monotonic attention mechanism, linking learners' future responses to assessment

questions with their past responses. Besides similarity between questions, attention weights also use exponential decay and context-aware relative distance measurements for computation. The Graph Knowledge Tracing (GKT) model (Nakagawa, Iwasawa, & Matsuo, 2019) is one of the earliest models to apply graph neural network models for knowledge tracing. It views a student's answer history as a three-part graph with student, question, and student answer sequence information. Although attention mechanisms and graph neural networks have shone in the KT field, they often do not take into account the impact of question hierarchy on knowledge states. **The dynamic tracking target hierarchy attention mechanism proposed in this paper can effectively capture the influence of question hierarchy difficulty on student target hierarchy.**

Recent studies have increasingly focused on the interpretability within the field of knowledge tracing. For instance, Wang et al. (2023) integrated the interpretability of prior educational knowledge with deep knowledge tracing through cognitive diagnosis. Meanwhile, Shen, Chen, et al. (2022) assessed the dynamic changes in students' knowledge states from the perspective of problem difficulty. Further, Chen, Liu, Huang, Liu, and Luo (2023) aimed to precisely simulate changes in students' knowledge states on a finer granularity level by learning problem-sensitive cognitive representations through modules centered around knowledge acquisition and problem-solving. Moreover, Shen, Chen, et al. (2022) advanced research on the interpretability of knowledge tracing by simulating the gains and forgetting of knowledge during the learning process to assess changes in students' knowledge states. **Our model enhances the interpretability of changes in student knowledge states from the perspective of the education target hierarchy.**

3. Preliminary

In this section, we provide formal definitions of knowledge tracing, KCs accuracy, and the frequency of occurrence of KCs. Additionally, we introduce some crucial embeddings in THKT.

3.1. Problem definition

In online intelligent tutoring systems, the set of students is defined as S, the set of questions as P, the set of KCs as KC, and the set of student response actions as R. The student set $S = \{s_1, s_2, \ldots, s_n\}$ comprises n different students, and the problem set $P = \{p_1, p_2, \ldots, p_m\}$ includes m different problem information. Each student can have N ($N \geq 1$) problem response records, and the same problem can be answered repeatedly. The KC set $KC = \{kc_1, kc_2, \ldots, kc_j\}$ contains j different KCs, with each problem possessing specific KCs. The response set $R = \{0, 1\}$ is a binary label related to the correctness of the answer to the problem (i.e., **0** indicates that the student answered the question incorrectly, and **1** indicates that the student answered the question correctly). Each student confirms their knowledge mastery from the response r ($r \in R$) corresponding to the problem by continuously answering different problems. During the practice process, the student's learning record can be recorded as $L = \{(p_1, kc_1, r_1), (p_2, kc_2, r_2), \ldots, (p_t, kc_t, r_t)\}$, where p_t is the problem answered by the student at time step t, kc_t is the KC corresponding to the problem answered by the student at time step t, a_t is the correct answer label related to the problem p_t answered by the student at time step t, and t is the length of the student's practice sequence. The definition of knowledge tracing is as follows:

Knowledge Tracing (KT): Given a student's exercise sequence $L = \{(p_1, kc_1, r_1), (p_2, kc_2, r_2), \ldots, (p_t, kc_t, r_t)\}$, the KT task is a dynamic process where the student's knowledge state changes over time and with learning. The task aims to determine their mastery of KCs based on their historical answering situations and further predict the student's performance on these KCs. Further, by modeling and analyzing the student's answering situations, personalized learning suggestions and guidance can be provided to maximize the student's learning efficiency.

3.2. KC hierarchy definition

In the large number of student response records, the accuracy and frequency of each KC are different (e.g., the accuracy and frequency of addition and subtraction are generally high, while those of logical reasoning are relatively low). This paper defines two aspects for each KCs: accuracy A_j and frequency F_j. In this study, we were inspired by Classical Test Theory (CTT) (DeVellis, 2006) and previous related research, so we adopted prior statistical methods to evaluate the accuracy and frequency of KCs, as shown below:

$$A_j = \frac{1}{N_j} \sum_{n=1}^{s_n} \{r_{nj} == 1\},$$
$$F_j = \frac{1}{max(J)} \sum_{n=1}^{s_n} f_{nj}, \tag{1}$$

where s_n represents the size of the student set, r_{nj} is the label indicating whether the nth student answered correctly on the jth KCs, and N_j is the total number of times the jth KCs appears. f_{nj} is the number of times the jth KCs appears for the nth student, and $max(J)$ refers to the maximum number of times any KCs appears.

KC Hierarchy: Hierarchical training is conducted based on the accuracy A_j and frequency F_j of KCs. The hierarchical structure can help teachers and students better understand the mastery of KCs, and make more targeted improvements.

3.3. KC hierarchy, question hierarchy, target hierarchy

In this paper, we categorize "Hierarchy" into three types: Knowledge Concepts Hierarchy (KCH), Question Hierarchy (QH), and Target Hierarchy (TH). KCH is a hierarchy at the knowledge concept level, derived based on the accuracy and frequency of KCs. Question Hierarchy (QH) built upon KCH, is a hierarchy at the problem dimension obtained by integrating problem information and student response for training. It reflects the hierarchy to which the problems practiced by the students should be attributed. On the other hand, Target Hierarchy (TH) is the current learning hierarchy of students, determined by deeply focusing on the QH involved in the students' exercise sequence. All three types of hierarchies uniformly adopt a six-hierarchy structural model.

3.4. Embeddings

Before delving into the overall structure of THKT, we can briefly outline the basic embeddings in THKT from the following two aspects, in order to better understand the internal operating mechanism of THKT.

3.4.1. Problem, KC and response embeddings

In the THKT model, the basic elements are Problem, Knowledge Concepts (KCs), and Response. Specifically, we utilize an embedding matrix $\mathbf{P} \in \mathbb{R}^{m \times d_p}$ (where d_p denotes the size of the embedding dimension) to represent all problems tackled by students. On the other hand, we employ an embedding matrix $\mathbf{KC} \in \mathbb{R}^{j \times d_k}$ (where d_k represents the size of the embedding dimension) to depict all KCs related to the problem IDs. Additionally, we use $\mathbf{R} \in \mathbb{R}^{2 \times d_r}$ (where d_r is the size of the embedding dimension) to signify the accuracy of student responses to different problem IDs (**1** signifies the problem was answered correctly, **0** suggests the problem was answered incorrectly).

3.4.2. KC accuracy and frequency embeddings

The accuracy A_j and frequency F_j are important variables for the KC hierarchy. Their original values are all decimals less than 1, which is inconvenient for embedding processing. For simple and efficient embedding of these two types of data, we first use $\lfloor 1000 * e^{A_j} \rfloor$ and $\lfloor 1000 * e^{F_j} \rfloor$ to scale them in a regular way. In addition, since the accuracy and frequency of some KCs do not appear in the prior

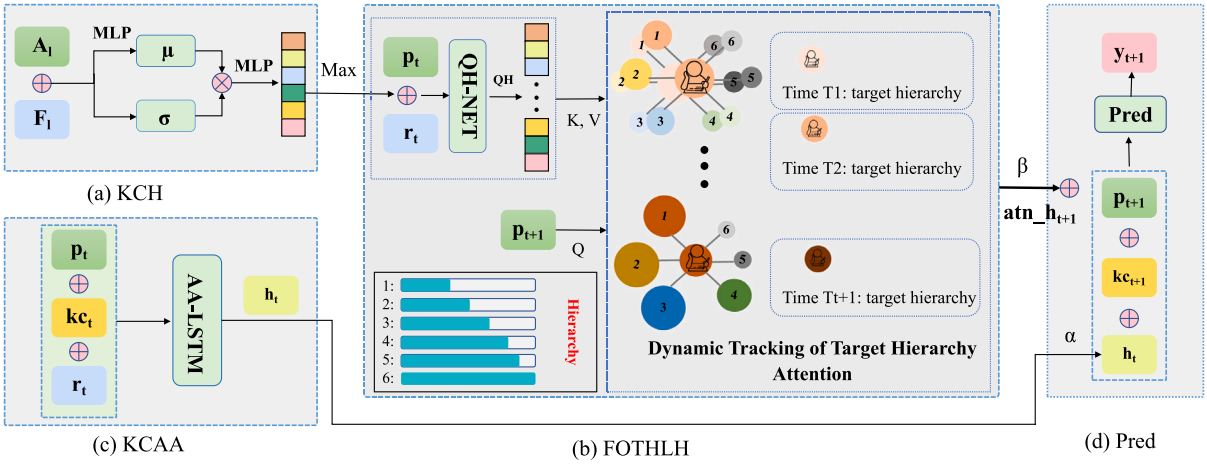


Fig. 2. The main structure of our THKT model is presented. We illustrate the processing flow of THKT at time step t. At this time step, the inputs are the accuracy embedding of KC A_j, frequency embedding F_j, problem embedding P_t, knowledge concept of the problem KC_t, student's response r_t, and knowledge state h_t. In addition, we can also predict the student's performance at time step $t+1$, denoted as y_{t+1}.

statistics, but they may appear in the future student practice process, we uniformly use $\lfloor 1000 * e^{0.5} \rfloor$ to replace them. It is important to note that although at the beginning of the learning cycle, the values of A_j and F_j are considered initial benchmarks based on prior statistics, our model allows these values to be dynamically updated throughout the learning process based on the actual performance of the students. This update mechanism aims to more accurately reflect the progress of students' learning and their mastery of knowledge. Consequently, we use $\mathbf{A_j} \in \mathbb{R}^{j \times d_a}$ (where d_a is the size of the embedding dimension) to represent the embeddings of the accuracy rates of all KCs. Similarly, $\mathbf{F_j} \in \mathbb{R}^{j \times d_f}$ (where d_f is the size of the embedding dimension) is used to represent the embeddings of the frequency of all KCs.

4. The THKT framework

In this section, we will detail the THKT framework proposed in this paper. The THKT framework consists of four parts, as shown in Fig. 2. We first construct a knowledge concepts hierarchy (KCH) network by conducting deep training based on the KC accuracy and frequency calculated from prior statistics, as shown in Fig. 2(a). Subsequently, we introduce problem IDs and responses from the student's answer sequence, and combine this with the KCH to design a question hierarchy (QH) discrimination network. Furthermore, based on learning history, we design a network (FOTHLH) that focuses on the target hierarchy (TH), as shown in Fig. 2(b). Further, through the joint learning of problem IDs, KCs, and responses, we construct a model for the student's KCs application ability (KCAA), as shown in Fig. 2(c). Finally, by integrating the learning of the target hierarchy and application ability of students in the FOTHLH and KCAA networks, we make the next prediction, as shown in Fig. 2(d). We will provide a more detailed introduction to these four modules below.

4.1. KCH: KC hierarchy

The educational hierarchy in which a student is located is extremely important for both teaching and learning. Clearly grasping and self-recognizing the current hierarchy is the inevitable path to implement precise teaching (Li, Rakovic, Poh, Gaševic, & Chen, 2022; Newman & Newman, 2022). However, most KT models only use KCs to represent problem information, rather than using the problems themselves, mainly because the large number of problems can easily cause sparsity issues (Chen et al., 2023). To avoid the poor effect of hierarchy due to insufficient information caused by directly calculating the sparse problem accuracy and frequency. In this paper, to address the issue of rigidity in the prior statistics of using knowledge concepts, we have adopted a deep learning approach to maintain the dynamism of the KC hierarchy. Specifically, we concatenate the accuracy rate A_j and frequency F_j of the KCs, and then enhance the data representation through a multi-layer perceptron (MLP), while using a gating mechanism to comprehensively consider the impact of the enhanced accuracy rate A_j and frequency F_j on hierarchy. Then, we categorize the KCs into six hierarchies, selecting the highest level as the hierarchy KCH_l for the KC at the current timestep. The specific calculation method of the KCH network is as follows:

$$\begin{aligned} x_l &= A_l \oplus F_l, \\ y_l &= \mu(W_l^1 x_l + b_l^1) \cdot \sigma(x_l), \\ KCH &= W_l^2 y_l + b_l^2, \\ KCH_l &= \max(\text{softmax}(\text{ReLU}(KCH))), \end{aligned} \tag{2}$$

where $\oplus$ is vector concatenation, $\mu(\cdot)$ and $\sigma(\cdot)$ are nonlinear activation functions $tanh$ and $sigmoid$, $W_l^1 \in \mathbb{R}^{(d_a+d_f)\times(d_a+d_f)}$ and $W_l^2 \in \mathbb{R}^{(d_a+d_f)\times n_{KCh}}$ are learnable weight parameters (n_{KCH} is the number of KC hierarchy), $b_l^1 \in \mathbb{R}^{(d_a+d_f)}$ and $b_l^2 \in \mathbb{R}^{n_{KCH}}$ are bias terms, and $max(\cdot)$ is the maximum value function.

4.2. FOTHLH: Focus on target hierarchy based on learning history

After obtaining the educational hierarchy of the KCs, the current hierarchy is still calculated through the KCH network based on the prior statistics of the KCs, which does not consider the individualized learning situation of the students. According to researches such as (Chen et al., 2023; Liu, Huang, et al., 2019; Yang et al., 2021), the problem has a significant impact on the judgment of the student's learning state. In order to more accurately consider the student's performance in a fine-grained manner, we need to consider the impact of the problem on the learning state during the student's exercise process. For example, if a student answers the KC (cube root) correctly, but immediately answers this KC incorrectly, if only considering from the perspective of the KC, it will give people a hard time understanding the student's behavior. However, if we consider the performance of the same KC under different problem IDs, we can have a good explanation of the student's knowledge mastery status (for example, the same KC belongs to different hierarchies in different problems). The KCH representation based on KCs is a statistical classification and does not have individualization.

In view of the above considerations, for different student answer sequences, we need to further bridge the specific performance of students on the same KC in different problems, and it is very necessary to introduce students' actual feedback on different problems. Specifically,

we need to focus more on the personalized learning history that students have achieved in the entire exercise process. Subsequently, based on this learning history, we assign an appropriate question hierarchy to this problem. In other words, if a student successfully answers a problem with a higher question hierarchy, we should consider that the student has a higher knowledge hierarchy and a better target hierarchy should be assigned to he/she. Similarly, if the student often answers incorrectly on problems with lower question hierarchies, then it should be considered that the student has a relatively lower target hierarchy.

In order to distinguish the student's hierarchy in a fine-grained manner, we propose the Focus on Target Hierarchy based on Learning History (FOTHLH) network to establish the relationship between the KCH where the student is currently located and the learning history experienced by the student. This network is mainly implemented in two steps: Firstly, the problem ID and the student's response to the question are introduced and incorporated with the generated KCH. Through the $QH-NET$ model, we obtain the student's knowledge mastery status and conduct an assessment of the question hierarchy (QH) at each time step. Secondly, pay attention to the relevance between QH and the student's target hierarchy, as it can have a significant impact on the student's learning target hierarchy.

To achieve a finer granularity of hierarchy for the extensive KCs-based KCH, we introduce the problem (p_t) from the student's answer sequence and the student's response (r_t) to that problem ID. Consequently, we obtain the representation of the student's knowledge state and the question hierarchy:

$$\begin{aligned} h_t^p &= LSTM(p_t \oplus KCH_l \oplus r_t), \\ QH_t &= \text{round}(5 \cdot \sigma(w^{QH} \cdot h_t^p) + 1), \end{aligned} \tag{3}$$

where $W^{QH} \in \mathbb{R}^{6\times seq}$, $h_t^p \in \mathbb{R}^{seq_l \times d}$ (seq_l is the length of the student's exercise sequence), and $LSTM$ is the Long Short-Term Memory network.

The student's knowledge status performance h_t^p around the problem can well obtain the student's knowledge performance at time step t, but it cannot deeply evaluate the correlation between the problem to be practiced next and the student's precursor knowledge and question hierarchy. Therefore, we use the attention mechanism for deeper focusing, and predict the target hierarchy at the next time step based on the student's performance on the historical question hierarchies. First, we use the following equation to obtain the query and key–value pairs:

$$Q = p_{t+1}W^Q, \quad K = QH_tW^K, \quad V = QH_tW^V, \tag{4}$$

where $W^Q, W^K, W^V \in \mathbb{R}^{d\times d}$ are the projection matrices for query, key and value respectively, which linearly project their respective vectors into different spaces. In order to make the next exercise related only to the question hierarchies that the student has studied, we use a lower triangular $mask$ matrix to calculate the attention mask, as shown in formula (5).

$$\text{mask} = \begin{bmatrix} -\infty & -\infty & -\infty & \cdots & -\infty \\ a_{21} & -\infty & -\infty & \cdots & -\infty \\ a_{31} & a_{32} & -\infty & \cdots & -\infty \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{t1} & a_{t2} & a_{t3} & \cdots & -\infty \end{bmatrix}, \tag{5}$$

the correlation between the prerequisite question hierarchy and the next exercise is calculated using attention weights, determining the target hierarchy the student should currently belong to. To more comprehensively seek attention weights, we use scaled dot product (Vaswani et al., 2017), defined as:

$$\begin{aligned} &\text{Atn}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}} + \text{mask}\right)V, \\ &atn_h_{t+1} = \text{cat}(\text{Atn}_1, \ldots, \text{Atn}_h)W^O, \\ &TH_{t+1} = \sigma(w^{TH} \cdot atn_h_{t+1}), \end{aligned} \tag{6}$$

where $W^O \in \mathbb{R}^{hd\times d}$, $W^{TH} \in \mathbb{R}^{6\times seq}$, $cat(\cdot)$ is a concatenation function, $atn_h_{t+1} \in \mathbb{R}^{seq_l \times d}$ is the correlation between the student's future target hierarchy and the learning history prior to time step t. TH_{t+1} is the target hierarchy of time step $t+1$.

4.3. KCAA: KC application ability

Simply obtaining the target hierarchy atn_h_{t+1} of the student's current learning history through the joint action of KCH and FOTHLH networks is not comprehensive enough for understanding the student's cognition. It is also necessary to understand the hierarchy of the student's specific application ability of knowledge to make a relatively target judgment on the student's knowledge state. In other words, the target hierarchy (TH) merely defines the learning level that the student has reached and does not accurately reflect their performance on specific questions. For instance, if two students both score 90, their target hierarchies might be assessed by the model as having reached the fifth hierarchy of competency. Despite their identical scores, the questions they answered correctly might not be exactly the same. Only by combining the specific questions that students answered correctly with their TH can we genuinely assess their knowledge mastery. In view of this, we combine the problem p_t encountered in the student's learning sequence, the KCs kc_t, and the student's response r_t to the problem for joint learning, to obtain the student's developing KC Application Ability (KCAA). Specifically, we use information fusion to enrich the student's learning features, and then judge the student's current application ability performance through an application ability learning layer, the formula is as follows:

$$h_t = LSTM(p_t \oplus kc_t \oplus r_t), \tag{7}$$

where $h_t \in \mathbb{R}^{d\times d}$ represents the manifestation of a student's application ability at time t.

4.4. Pred: Prediction net

Since we have obtained the knowledge application ability h_t that the student possesses at the current time step t through the KCAA network, and at the same time, we have also obtained the correlation atn_h_{t+1} between the student's target hierarchy and the student's learning history through the FOTHLH network, we can predict the student's future performance at time step $t+1$. In the THKT framework, we integrate the application ability at time step t, the problem p_{t+1} and the KC kc_{t+1} at time step $t+1$ to enhance the representation of the student's application ability expansion using the learned knowledge. At the same time, combined with the learning history atn_h_{t+1} to varying degrees, we jointly predict the performance of the student at time $t+1$, the formula is as follows:

$$\begin{aligned} y_t &= p_{t+1} \oplus kc_{t+1} \oplus h_t, \\ y_{t+1} &= \sigma(w_t^1(\alpha y_t \oplus \beta atn_h_{t+1}) + b_t^1), \end{aligned} \tag{8}$$

where $w_t^1 \in \mathbb{R}^{(d_p+d_k+d+d)\times 1}$ is a learnable weight parameter, $b_t^1 \in \mathbb{R}^1$ is a bias term, and $\sigma(\cdot)$ is the nonlinear activation function.

4.5. Objective function

In order to train all the parameters in the THKT framework, we use the cross-entropy log loss between the framework's predicted value y and the actual correctness of the student's response to the problem r as the target function:

$$\mathcal{L}(\theta) = -\sum_{t=1}^{T}(r_t \log(y_t) + (1-r_t)\log(1-y_t)) + \lambda_\theta \|\theta\|^2, \tag{9}$$

where θ represents all the parameters of THKT, and λ_θ represents the regularization hyperparameters. We use the Adam optimizer (Zhang, 2018) to minimize the target function on small batches. The details of other settings will be elaborated in more detail in the experimental section.

Table 1
Statistics of all datasets.

Statistics \ Datasets	ASSIST2009	ASSISTchall	EduNet-KT1
Students	4217	1709	784,309
Problems	17,751	3162	13,169
Concepts	123	102	188

5. Experiment

In this section, we begin by presenting the real-world datasets utilized in our experiment, followed by a detailed explanation of the training process and the baseline model. Subsequently, we compare the forecasts of student future performance as predicted by both the baseline model and the model introduced in this study. Furthermore, to more effectively illustrate the interpretability of the THKT model and the efficacy of the proposed educational goal hierarchy, we carry out a series of experiments focusing on the following eight dimensions:

- (1) The THKT model furnishes forecasts regarding students' prospective academic performance.
- (2) The impact of dynamic tracking of the target hierarchy on the positioning of the target hierarchy.
- (3) The adaptability of the pre-established statistical hierarchy of Knowledge concepts (KCs).
- (4) Comparing the impact on performance of static hierarchy division based on the normal distribution.
- (5) Exploring the differences in how different students perceive the same question.
- (6) A comparative study of question hierarchies from model learning and expert annotations.
- (7) The influence of various elements in THKT.
- (8) How target hierarchy affects the performance of the THKT model.

5.1. Datasets

We use three different real-world public datasets to evaluate the effectiveness of THKT: (1) ASSIST2009 (2) ASSISTchall and (3) EdNet-KT1. The statistical data of the datasets are shown in Table 1. In this study, the exercise sequence lengths for the ASSIST2009, ASSISTchall, and EduNet datasets are set to 50, 500, and 120, respectively. If a sequence length exceeds this setting, it will be truncated into multiple virtual students. Conversely, if a sequence length is shorter, the remaining part will be padded with zeros.

- ASSIST2009[1] is a dataset used for educational data mining and learning analytics. It was created by Professor Neil Heffernan and his team at Worcester Polytechnic Institute with the goal of assisting researchers and educators in understanding students' behaviors and performances in online learning environments. The dataset collects online learning data from high school students in the United States, including students' response records, exercise content, and personal characteristics. These data can be used to analyze students' learning patterns, evaluate their knowledge hierarchies, and predict their learning outcomes.
- ASSISTchall[2] is a challenging dataset designed for assessing students' abilities. It was led by Professor Neil Heffernan and Professor Ryan Baker, with the aim of examining students' capabilities and question-solving skills across various knowledge domains.
- EdNet-KT1[3] is a dataset collected from 784,309 students at Santa since 2017. The dataset contains a large amount of educational text. To our knowledge, in terms of student population, interactions, and types of interactions, this is the largest publicly available education dataset. It consists of four different levels of datasets, and this study utilizes the KT1 version.

5.2. Experimental setup

In our experiments, we first removed records with empty $\epsilon problem\ ID\epsilon$ or $\epsilon kc\epsilon$ fields and sorted the remaining records based on the corresponding order fields in each dataset. Specifically, for the ASSIST2009 dataset, we sorted the records in ascending order of the $\epsilon order_id\epsilon$ field. For the ASSISTchall and EdNet-KT1 datasets, we sorted the records based on the $\epsilon start_time\epsilon$ field. During the training phase, we randomly initialized all parameters. For each dataset, we set the random seed to 1025 and distributed the data as follows: 60% to the training set, 20% to the validation set, and the remaining 20% to the test set. To streamline the model design, all dimensional parameters, including d_p, d_k, d_r, d_a, d_f, and d, were uniformly set to 256 in our THKT implementation. We established the batch size for all experiments at 32 and utilized 8 heads for the multi-head attention mechanism. All experiments were carried out on a 64-bit Ubuntu 20.04.5 LTS server outfitted with an Intel (R) Xeon (R) Gold 5218 CPU @ 2.30 GHz and a single 32 GB Tesla-V100 GPU. All code implementations were executed utilizing the PyTorch framework.

5.3. Baselines

To evaluate the effectiveness of THKT, we compare it with sixteen different baseline methods. To ensure a fair comparison, they used the same training, validation, and test sets as THKT. All these methods were adjusted to ensure the best performance from the implementation. To meet the comparison needs of different implementations of target hierarchies, we divide the baseline models into two categories. Specifically, one category is the prediction of student knowledge states involving only the KCs kc_t and response r_t. The other category includes the problem ID p_t, kc_t, and response r_t participating in the prediction of student knowledge states.

5.3.1. Without problem ID

- **DKT** (Piech et al., 2015) as the first knowledge tracing (KT) model to incorporate deep learning, relies on Recurrent Neural Networks (RNN) or Long Short-Term Memory (LSTM) to estimate students' mastery of knowledge.
- **DKVMN** (Zhang et al., 2017) enhances the capabilities of traditional memory-augmented neural networks by using a key–value structured memory, where the key component is used to address relevant memory slots and the value component is used to aggregate content. This allows for a more flexible and expressive representation of student knowledge.
- **DKT+** (Yeung & Yeung, 2018) identifies two primary challenges in the original DKT model: Firstly, DKT is incapable of reconstructing observed input learning sequences; Secondly, DKT demonstrates inconsistency in predicting students' future performance across time steps. In order to address these issues, DKT+ introduces several regularization factors.
- **DKT-DSC** (Minn et al., 2018) employs the K-means clustering algorithm to group students with similar ability hierarchies into the same cluster at each time interval. After learning the centroids of all K clusters, each student is assigned to the nearest cluster for subsequent prediction.

[1] https://sites.google.com/site/assistmentsdata/home/2009-2010-assistment-data
[2] https://sites.google.com/view/assistmentsdatamining
[3] https://github.com/bigdata-ustc/EduData?tab=readme-ov-file

- **SAKT** (Pandey & Karypis, 2019) model employs a self-attention mechanism to gauge the importance of each past interaction. This means that the model can learn to focus on the most relevant past interactions when making predictions, rather than treating all past interactions equally.
- **CKT** (Shen et al., 2020) designed hierarchical convolutional layers to extract their learning rates based on students' continuous learning interactions, automatically learning meaningful exercise embeddings.
- **JKT** (Song et al., 2021) models the multidimensional relationships of "exercise to exercise" and "concept to concept" as graphs, and integrates them with the "exercise to concept" relationship. This not only enables the establishment of connections between cross-concept exercises, but also aids in capturing high-level semantic information, enhancing the interpretability of the model.
- **LPKT** (Shen et al., 2021) monitors students' knowledge states by directly modeling their learning process. It specifically designs a Learning gain module to simulate students' ability to absorb knowledge, and a forgetting effect module to simulate the decline of students' knowledge over time.
- **DNeuralCDM** (Wang et al., 2023) integrates the interpretability of educational prior knowledge from cognitive diagnosis into deep learning-based knowledge tracing methods, and makes predictions about the student's knowledge state.

5.3.2. With problem ID

- **EKT** (Liu, Huang, et al., 2019) takes into account the relevance of each exercise to all KCs, based on the semantic information of the exercises as input.
- **GIKT** (Yang et al., 2021) utilizes Graph Convolutional Networks (GCN) to fully integrate the relevance of problems and skills through embedding propagation.
- **MF-DAKT** (Zhang et al., 2021) augments the representation of problems and employs various factors, utilizing a dual attention mechanism to accurately model the progression of student learning.
- **DIMKT** (Shen, Huang, et al., 2022) explicitly includes difficulty hierarchies in the problem representation, and designs a three-stage adaptive sequence neural network to establish the relationship between a student's knowledge state and problem difficulty during the practice process.
- **QRAKT** (Li & Wang, 2023) is a Calibrated Q-matrix Relationship Attention Knowledge Tracing model, which is developed using a Q-matrix calibration method based on hierarchical knowledge hierarchies.
- **QIKT** (Chen et al., 2023) method explicitly models the changes in a student's knowledge state at a fine-grained level by learning problem-sensitive cognitive representations from a problem-centered knowledge acquisition module and a question-solving module.
- **TCKT** (Huang et al., 2024) introduces a causal self-attention mechanism based on the theory of front-door adjustment, which improves the interactive representation and reduces prediction errors caused by dataset bias.

5.4. Future performance prediction

While the core goal of THKT is to establish a reasonable learning history model based on the student's learning process, its further goal is to develop personalized learning progress plans for each target hierarchy. Although our goal is not purely to predict students' future performance, the prediction of students' future performance is still one of the important indicators for evaluating the effectiveness of KT. Usually, the higher the accuracy of predicting students' future performance, the more precise the evaluation of their knowledge state. To evaluate the prediction of students' future performance by the THKT model proposed in this paper, we compare this model with the baseline models mentioned above. Specifically, for each student's exercise sequence, except for the first question that does not participate in future predictions, all other questions have participated in the performance prediction at the corresponding time. For a more comprehensive comparison, this study employs two metrics, Accuracy (ACC) and Area Under the Curve (AUC), in the comparative experiments.

The comparison experiment results of THKT and other models on three benchmark datasets are shown in Table 2, from which we can find several important observations. First of all, THKT's ACC performance on all datasets is better than other baseline models. Although the model's AUC performance on the ASSIST2009 dataset is not optimal, it has also achieved a sub-optimal performance of 83.57%, which is not much different from the best we know, and the AUC has reached the best on the remaining two datasets. The good performance of THKT indicates that it is in line with the development of education and necessary to divide students into different target hierarchies based on their learning history. Secondly, the performance of models that use *problems* is not always better than that of models without *problems*. This is mainly because the *problem* in the education dataset has always had a long tail problem. Determining how to effectively utilize problem IDs to enhance the prediction of students' future performance remains an intriguing and worthwhile area of exploration.

Further observation, existing KT models with *problems* (e.g., DIMKT) perform better on average on two main indicators when the student exercise sequence is relatively long (e.g., ASSISTchall) than KT models without *problems* (e.g., DKT, SAKT). This also shows that by paying attention to the learning sequence for a long time, more personalized knowledge of students can be obtained, including our model THKT which has achieved the best performance on both indicators on long sequences, further proving that a longer sequence may have a more accurate grasp of the student's learning history, and it is realistic to divide the student's current learning hierarchy based on the learning history. Following this, when compared with the group KT model with similar practices (e.g., DKT-DSC), our model is different from the DKT-DSC model which dynamically groups students based on similarity of ability. We adopt a method where each student builds their own independent learning history model, which is more personalized to some extent, and therefore performs better than this model. Finally, compared with LPKT, which simulates the student learning process, DIMKT, which enhances difficulty embedding, and MF-DAKT, which divides difficulty levels, our model has not shown a huge advantage on these data, which encourages us to carefully analyze the advantages of each model and further explore in the future, striving for greater breakthroughs.

5.5. Dynamic tracking of target hierarchy

The impact of the dynamic target hierarchy attention module on students' target hierarchy (TH) is significant. As demonstrated in Fig. 3, the data in the figure suggest that a student's TH dynamically adjusts based on their correct or incorrect responses during exercises. For correctly answered question hierarchy (QH), TH typically allocates a larger attention weight, while for incorrectly answered QH, a lower attention weight is assigned. This adjustment mechanism allows the model to precisely identify the student's TH, and also reflects the module's effective simulation of human learning processes: correct responses have a substantial positive impact, while incorrect responses can, to a certain extent, represent a student's skill level.

An in-depth analysis of Fig. 3(a) discloses that student $S1$ initially answered questions at $QH = 6$ and $QH = 4$ incorrectly. These two questions failed to accurately determine the question hierarchy that the student could handle, thus setting S1's TH to 1. However, as the exercises progressed, $S1$ gradually managed to competently handle questions at $QH \leq 4$. Concurrently, S1's TH began to monitor these

Table 2
Performance comparison (%) of different methods on the ASSIST2009, ASSISTchall, and EdNet-KT1 datasets, the best results are indicated in bold, while the second best results are underlined.

Category	Methods \ Datasets	ASSIST2009		ASSISTchall		EdNet-KT1	
		ACC	AUC	ACC	AUC	ACC	AUC
Without problem ID	DKT (Piech et al., 2015)	72.10	74.40	70.57	72.63	65.60	69.20
	DKVMN (Zhang et al., 2017)	80.16	81.57	68.42	71.08	68.43	67.41
	DKT+ (Yeung & Yeung, 2018)	73.51	75.83	74.17	71.94	69.24	68.59
	DKT-DSC (Minn et al., 2018)	73.34	73.89	49.57	44.18	67.99	66.94
	SAKT (Pandey & Karypis, 2019)	74.78	75.20	66.94	66.05	68.62	67.94
	CKT (Shen et al., 2020)	77.65	82.45	69.24	72.63	68.71	68.11
	JKT (Song et al., 2021)	75.45	79.84	73.97	75.18	70.45	68.71
	LPKT (Shen et al., 2021)	81.77	82.12	74.24	80.08	73.00	77.21
	DNeuralCDM (Wang et al., 2023)	74.82	78.39	70.11	70.24	69.99	68.25
With problem ID	EKT (Liu, Huang, et al., 2019)	70.02	75.40	72.70	72.00	68.12	62.26
	GIKT (Yang et al., 2021)	77.78	78.58	74.06	74.32	71.26	75.25
	MF-DAKT (Zhang et al., 2021)	79.31	**84.15**	77.35	77.46	71.26	76.97
	DIMKT (Shen, Huang, et al., 2022)	81.03	81.89	78.22	76.09	71.05	72.49
	QIKT (Chen et al., 2023)	79.56	78.78	72.64	73.58	64.59	64.26
	QRAKT (Li & Wang, 2023)	74.20	73.59	72.36	72.06	65.88	63.25
	TCKT (Huang et al., 2024)	80.11	79.64	72.34	78.75	72.76	75.49
Ours	THKT	**82.56**	83.57	**78.26**	**80.35**	**73.18**	**77.86**

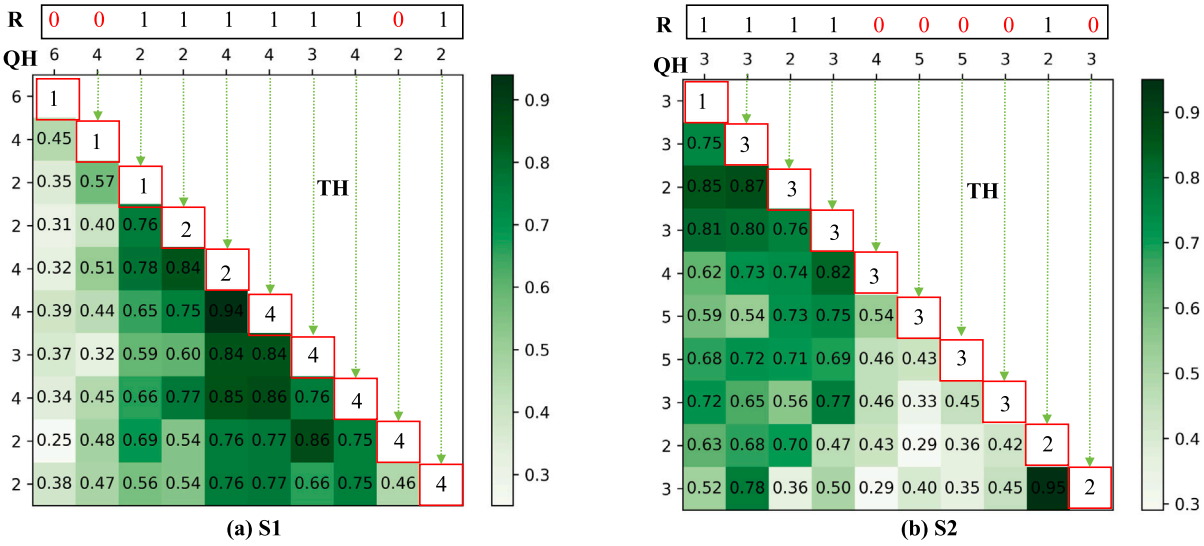


Fig. 3. The schematic diagram of the impact of dynamic tracking of the target hierarchy on the positioning of the student's target hierarchy in dataset ASSISTchall. In which, R represents the genuine responses of students to the questions, 1 represents correct answers, and 0 represents incorrect answers. QH represents the question hierarchy, and TH represents the student's target hierarchy. The data indicated by the dashed arrow in the red box represents the current target hierarchy calculated at time step t based on the performance of the previous t-1 question hierarchies. This sequence is the student's exercise sequence, only retaining the question hierarchy of different problems, omitting the problem ids and KCs.

changes, assigning greater weight to these question hierarchies, thereby temporarily positioning S1's TH at 4 .

Further inspection of Fig. 3(b) reveals that student $S2$ performed exceptionally well at the onset of the exercises, correctly answering all questions at $QH \leq 3$. During this phase, S2's TH dynamically monitored these changes, temporarily setting its TH to 3 . However, $S2$ incorrectly answered subsequent questions at $QH = 4$ and $QH = 5$. At this juncture, the attention weight of TH was still primarily focused on questions at $QH \leq 3$, assigning lower weights to incorrectly answered questions, and S2's TH remained at 3 . A more detailed analysis reveals that when $S2$ incorrectly answered a question at $QH = 3$ again, its TH was temporarily set to 2. This suggests that TH's attention can effectively monitor a student's incorrect performance on questions, and these errors also serve a punitive role in altering a student's target hierarchy.

This experiment conclusively demonstrates that students' target hierarchy attention can effectively update its own weight dynamically as students delve deeper into the exercises, thereby influencing students' target hierarchy.

5.6. Adaptability of hierarchy

Given the diverse input types encompassed by numerous KT models, we have conducted a classification evaluation in this paper on whether the proposed hierarchy possesses a certain degree of adaptability. The first category pertains to the effects of integrating the KCH component into models that do not include problem information input. The second category examines the impact of incorporating the FOTHLH component into models that already include problem information input.

The core goal of THKT is to have a clear understanding of the educational target hierarchy at which a student is currently at. In order to verify the rationality and interpretability of the hierarchical concepts we propose, we grafted the two hierarchical ideas of KCH and TH onto four KT models on the EdNet-KT1 dataset, respectively. The comparison results are shown in Table 3. Through this table, we can clearly observe

Table 3
Comparison of AUC performance by transplanting the KCH component and the FOTHLH component into the corresponding modules, respectively. Where grafting the FOTHLH module means that the KCH module also participates in the model learning simultaneously.

Methods	+ Component	EdNet-KT1		
		Original	After Grafting	Gain
DKT	KCH	69.20	71.35	+2.15
SAKT	KCH	67.94	70.07	+2.13
QIKT	FOTHLH	64.26	68.42	+4.16
QRAKT	FOTHLH	63.25	68.59	+ 5.34

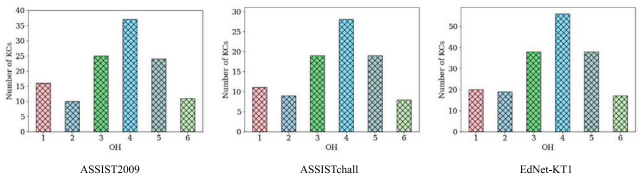


Fig. 4. Static hierarchy division based on the normal distribution of KCs in three datasets.

that the method of direct grafting of the hierarchy has a good effect on the original model. Specifically, for DKT and SAKT, these two KT models did not allow the problem information to participate in model learning. We only grafted the KCH component, the effect increased by more than 2.0% compared to the original model, which further illustrates that it is reasonable to model students at different hierarchies based on educational rules. This also indicates that the hierarchical division based solely on KCs can have a positive effect on the model.

Looking at the two KT models of QIKT and QRAKT that contain problem information, the use of the FOTHLH proposed in this paper, which is a more complete target hierarchy identification based on learning history, has achieved a gain of more than 4.0% in predicting future student performance. This also shows that the method of TH based on each student's own learning history has better performance.

Through this experiment, we have verified that both the KCH based on KCs and the complete concept of TH proposed in this paper have a certain adaptability and can be well applied to other models. This further demonstrates the rationality and a certain degree of universality of the target hierarchy. This also proves the presence of hierarchical in the KT datasets.

5.7. Static hierarchy based on prior KCs

The target hierarchical method used in the THKT model is a dynamic hierarchical division based on the accuracy and frequency of KCs, combined with the student's learning history under the premise of prior statistics. In educational testing (including exams), the normal distribution probability model is often used to estimate and calculate some issues (Salas-Rueda, 2019). According to educational theory, well-designed questions should basically conform to a normal distribution.

In summary, we design a static hierarchical algorithm SHM based on the accuracy and frequency of KCs. The process of this algorithm is shown in Algorithm 1. Firstly, we calculate the accuracy $Accuracy_i$ and frequency $Frequency_i$ of all KCs that appear in the training set, and then calculate $C_{AF_i} = Accuracy_i * Frequency_i$ to get its importance index for each KCs. Secondly, we sort each KCs's C_{AF_i} in ascending order, and use the percentile method to split the sorted $Sort_C_{AF}$, obtaining the corresponding position's C_{AF} value sp. Thirdly, we design a discriminative logic function $CalTH(C_{AF_i})$ to divide the hierarchy of KCs. Lastly, we return the KCs contained in each target hierarchy.

The setting of percentiles is roughly divided based on the theory of "large in the middle and small at both ends" often used in educational testing (Ehrlich et al., 2019). To verify the reasonableness of the division, we counted the KCs in three datasets divided by the ratio of $sp \in \{0.1, 0.3, 0.6, 0.8, 0.9\}$, as shown in Fig. 4.

Algorithm 1 Static Hierarchical Method

procedure SHM($Accuracy, Frequency$)
  Input: Accuracy and Frequency
  Output: The QH to which the KC belongs
  while i *in* KCs **do** ▷ Traverse All the KCs
    $C_{AF_i} \leftarrow Accuracy_i * Frequency_i$
  end while
  $Sort_C_{AF} \leftarrow Sort(C_{AF})$ *in ascending order*
  $sp \leftarrow [0.1, 0.3, 0.6, 0.8, 0.9]$, $TH_{1-6} \leftarrow [] \cdots []$
  $l_kc \leftarrow len(Sort_C_{AF})$, $sp1 \cdots 5 \leftarrow l_kc * sp[0 \cdots 5]$
  function CALTH(C_{AF_i})
    if $C_{AF_i} < sp1$ **then return** 6
    else if $sp1 \leq C_{AF_i} < sp2$ **then return** 5
    else if $sp2 \leq C_{AF_i} < sp3$ **then return** 4
    else if $sp3 \leq C_{AF_i} < sp4$ **then return** 3
    else if $sp4 \leq C_{AF_i} < sp5$ **then return** 2
    else return 1
    end if
  end function
  while C_{AF_i} *in* $Sort_C_{AF}$ **do**
    $TH_{CalTH(C_{AF_i})}.append(KC_i)$
  end while
  return QH ▷ KCs Hierarchy
end procedure

From the figure, it can be seen that the static goal stratification of the KCs in these three datasets basically satisfies the goal of a normal distribution-like "large in the middle, small at both ends", which is consistent with the dynamic goal stratification proposed in this paper based on student learning history from the principle of education.

We use the static hierarchical method to test by replacing the original KCH hierarchical results with the stratification results on three datasets. The test results are shown in Fig. 5. Through this figure, it can be clearly observed that for most of the baseline models participating in the comparison, the performance of the KCH component based on prior statistics and trainable on ACC and AUC is better than the SHM component. Further observation found that no matter which stratification method is used, the effect of the model using hierarchy is better than the effect of not using hierarchy. This once again proves that hierarchy plays a positive role in controlling the knowledge state of students. However, at the same time, we also found that in the DKT-DSC model, the method of using the SHM component is better than using the KCH component in terms of ACC performance. The possible reason is that this model itself dynamically groups students, and now adding a prior static KCs stratification allows the model to learn more knowledge.

5.8. Visualization of perceptual differences

Through the aforementioned experiments, we have explored the impact of three distinct hierarchical structures, KCH, SMH and $FOTHLH$, when used as target hierarchies on the model. Following this, we will conduct a more intuitive attention visualization experiment using question hierarchies. To clearly demonstrate the changes in the question hierarchy, we have taken the example of a student with a lower target hierarchy, with the comparative visualization shown in Fig. 6.

Upon analyzing Fig. 6, we first observe the student's performance to be quite poor, with most questions answered incorrectly. Upon further observation, we find that the hierarchical methods of SHM and KCH indicate that the questions practiced by the student are mostly around $QH = 3$. However, the student's responses are mostly incorrect, indicating that the student's hierarchy is significantly lower than the

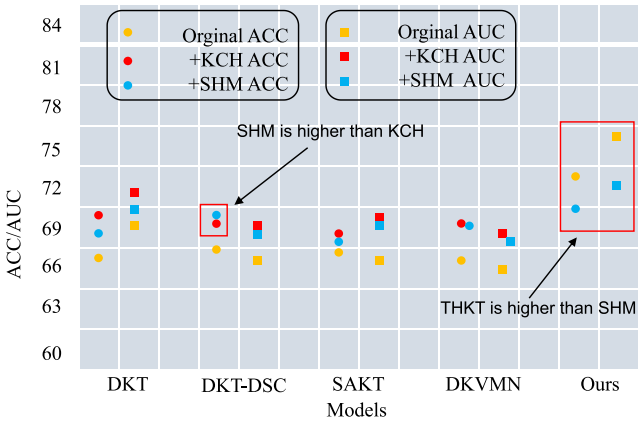


Fig. 5. Comparing the ACC and AUC effects after replacing the KCH component with the SHM component on the EdNet-KT1 dataset.

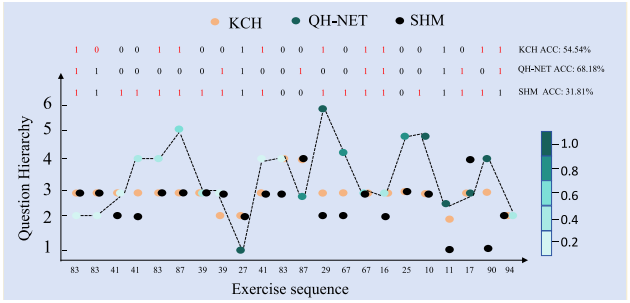


Fig. 6. The THKT model provides a dynamic visual comparison of three question hierarchy division methods on the EdNet-KT1 dataset. In this figure, the color of the QH-NET method continuously changes, with darker shades indicating a higher probability of being assigned to that question hierarchy. A higher probability suggests that the model is more confident in the current question hierarchy definition. A value of 0 indicates that the model predicts the student will not be able to answer correctly, while a value of 1 indicates that the model predicts the student will be able to answer correctly. Red denotes instances where the model's prediction does not align with the student's actual performance, indicating a prediction error.

average. Among these, KCH is used as the question hierarchy to participate in the model prediction, achieving an accuracy of 54.54%, while SHM is used as the question hierarchy to participate in the model prediction, achieving an accuracy of only 31.81%. The comparison between SHM and KCH indicates that for students whose target hierarchy is significantly lower than the average, the SHM question hierarchy judgment for prediction is disastrous, with an extremely low accuracy.

When we further observe the impact of the question hierarchy in $QH-NET$ after introducing the attention mechanism, we can see that the hierarchy fluctuates greatly, and most hierarchies are higher than the other two methods. This indicates that $QH-NET$ is more intelligent in judging the question hierarchy, and can appropriately increase the question hierarchy in combination with the student's poor performance to improve prediction accuracy. These questions are at a medium hierarchy (Applying) for most students, but due to the poor performance of the student in this practice, he feels that the question hierarchy is higher and cannot effectively answer. Therefore, the model will judge these questions to be at a higher hierarchy for the student based on his personalized learning history. The experiment proves the effectiveness of the $QH-NET$ model we proposed, which achieved the best prediction accuracy of 68.17% among the three methods. It also validates that different students perceive the same problem differently, and should be assigned different question hierarchies.

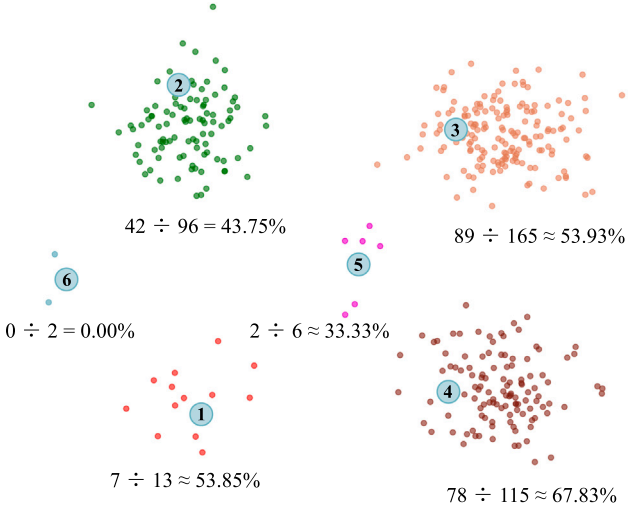


Fig. 7. The comparison between the question hierarchy (QH) calculated by our model and those annotated by experts according to Bloom's theory, where the number of questions annotated by experts at that hierarchy serves as the divisor, and the number of questions that match serves as the dividend.

5.9. hierarchical comparative study

This study classifies educational objectives into six hierarchies, inspired by Bloom's Taxonomy of Educational Objectives. Our purpose is not to precisely replicate Bloom's classification framework, but to explore whether our model can match these hierarchy classifications to some extent. For this purpose, we conducted a fundamental comparative study. Given the lack of explicit hierarchy annotations in the questions, we enlisted the help of two middle school mathematics teachers and two students to randomly select exercise data for 10 students from the ASSISTchall dataset. Following that, we chose 20 exercise questions from each academic year of 2004–2005 and 2005–2006 for expert annotation, totaling 400 exercise questions annotated. Out of these, 397 were effectively annotated.

As illustrated in Fig. 7, a detailed comparative analysis was conducted on 397 problems annotated by experts. The hierarchical results primarily focused on the second, third, and fourth hierarchies, with only two problems annotated at the highest hierarchy. This outcome generally aligns with our expectations regarding the hierarchical structure of the complexity of middle school mathematics problems. Further analysis revealed a partial mismatch between the hierarchical structure we proposed and the one determined through expert annotation. The highest match rate occurred at the fourth hierarchy, with a match rate of only 67.83%. However, considering the close relationship among the second, third, and fourth hierarchies, it may be challenging for expert annotations to precisely distinguish between these hierarchies. If we do not separately differentiate the second, third, and fourth hierarchies but instead combine them into a larger hierarchy, we find that the model's match rate with expert annotations can reach 81.37%. The match rate is defined as $\mathrm{MR}^h = \left(\frac{M^h}{T^h}\right) \times 100\%$, where MR^h is the match rate at the hth hierarchy, M^h is the number of matching questions at the hth hierarchy, and T^h is the total number of questions annotated by experts at the hth hierarchy. This discovery reveals an interesting phenomenon: although our classification method was inspired by Bloom's taxonomy, the actual correspondence did not fully align.

Despite these discrepancies, they do not detract from the primary focus of our research. Our main objective is to identify different question hierarchies through model analysis and, based on students' responses to these varied question hierarchies, to assess their learning objective hierarchies, thereby providing guidance for educational practice. The value of this research lies in its ability to clearly capture the hierarchical



Fig. 8. This is a demonstration of how we interpret future predictions by evaluating the target hierarchies in the learning sequences of two students using the EdNet-KT1 dataset. We have standardized the range of accuracy and frequency to be between 0 and 1, where a larger value signifies higher accuracy and frequency. In this context, QH stands for question hierarchy, and within TH, a darker color indicates a higher probability of assigning the student to that particular level at a given moment. $Accuracy of the pred$ represents the correctness of the THKT model's prediction for the exercise question at time t. The ✓indicates that the prediction matches the student's actual answer, while the × signifies that the prediction contradicts the student's actual response.

structure of student learning and apply this structure within the domain of knowledge tracing, offering a deeper perspective on understanding student learning progression.

5.10. Ablation study

In general, an ablation study in the context of machine learning models is a process to measure the impact of a specific component or feature by removing it from the model and comparing the performance of the full model and the ablated model. This process helps to understand the contribution of each component or feature to the performance of the model.

- THKT w/o KCH, which refers to THKT without considering the KCH based on prior statistics.
- THKT w/o FOTHLH, which refers to THKT not considering the component FOTHLH, and also not considering the KCH component.
- THKT w/o KCAA, which refers to THKT not considering the KCAA component of KCs application ability.
- THKT w/o problems, which refers to THKT not involving problem information.

The experimental results of the above variant models are shown in Table 4. From this table, it can be clearly observed that removing the FOTHLH module, that is, not considering the impact of hierarchy on students' future performance at all, the model degradation is more serious, which is one of the reasons why we propose the hierarchical concept. Further observation reveals that the THKT w/o KCAA, which only uses the hierarchy to predict student performance without considering student's application ability, also performs better than some previous models. This also indicates that a good target hierarchy has a good guiding significance for students' practice process. Finally, the THKT w/o problems, which does not bring the problem information into the model, shows that the loss of the main dependency of the personalized learning history has a larger negative impact on the model and cannot fully play the role of target hierarchy.

Table 4
The impact of each component on the THKT model.

Model	ASSIST2009		ASSISTchall		EdNet-KT1	
	ACC	AUC	ACC	AUC	ACC	AUC
THKT	**82.56**	**83.57**	**78.26**	**80.35**	**73.18**	**77.86**
THKT w/o KCH	74.65	75.42	72.64	73.95	67.80	70.12
THKT w/o FOTHLH	73.20	74.59	70.99	72.84	66.72	69.55
THKT w/o KCAA	78.56	76.22	73.49	73.97	72.58	72.29
THKT w/o problems	76.24	80.13	72.28	74.52	75.23	70.65

These findings underscore the significance of each component within the THKT model. The FOTHLH module and problem input play a pivotal role in capturing the individual learning history, thereby personalizing the learning trajectory. Conversely, the KCAA contributes towards a more comprehensive understanding of the student's application ability. This analysis could serve as a guide for future enhancements and adaptations to the THKT model or models of a similar nature.

5.11. Visualizing the impact of TH on future performance

The fine-grained target hierarchy division for students' personalized learning history plays a crucial role in the THKT model. This is demonstrated by describing the future performance of two students on the EdNet-KT1 dataset.

In Fig. 8, we can first observe that the THKT model has excellent predictive performance. Moreover, we can also find that the THKT model has good interpretability for whether the students will succeed or fail in their future practice.

For example, for student s_1, it can be seen that the hierarchies of the problems that the student can basically answer are all at $QH \leq 4$. For problems **1005** and **987**, which correspond to KCs **10** and **25** respectively, these two KCs have an $QH = 3$ in the current problem as determined by model training, and the student can answer them correctly. However, for the same KCs in different problems **507** and **761**, the $QH = 5$ for these two KCs. Although they are the same KCs, the different problem contexts may result in different hierarchies of

the KCs. Combined with the student's consistent learning history, the student's best performance is to answer questions with $QH = 4$. The student's target hierarchy should be set to $TH = 4$ at this stage. Thus, the student is unable to accurately answer questions at hierarchy level 4 and beyond, confirming the correctness of the model's prediction.

Further, it can be found that for the first KC of **19** practiced by the student, because the hierarchy of this KC combined with the problem is relatively low, the student can correctly answer this question, and the model also gives the correct prediction.

Similarly, for student s_2, the problems that the student can answer are all at a higher hierarchy. For example, for KC of **103** in problem **359** with $QH = 5$, even though this is the student's first time encountering this KC, the final prediction is consistent with their future performance. This is mainly due to the student's good performance in the early stages, and their target hierarchy has already been determined as $TH = 5$. In light of this, it can be predicted that the student will be able to correctly answer these types of questions. This once again verifies the importance and interpretability of the target hierarchy idea used in this paper.

However, it can also be found that the model cannot make correct predictions for potential psychological fluctuations. For example, for problems **8465** and **46**, the problem hierarchy is lower, and the model predicts based on the student's consistent performance that these types of problems can be easily solved, but the student answered them incorrectly. The possible reason is that the student was too relaxed or overthinking.

This experiment demonstrates the good interpretability of the THKT model, further affirming the rationality and necessity of the Target Hierarchy-guided Knowledge Tracing proposed in this paper. Meanwhile, this analysis highlights the strengths of the THKT model in capturing the complexity of learning processes and its limitations in accounting for psychological factors. Future work could explore ways to incorporate such factors into the model.

6. Conclusions and limitations

In this paper, we first analyze the necessity and significant impact of classifying problems by question hierarchies, and then dynamically tracking students' target hierarchies based on their feedback to different question Hierarchies. Then, we proposed a novel ***T**arget **H**ierarchy-guided **K**nowledge **T**racing* (***THKT***) model, which measures the effect of the target hierarchy and improves KT performance by establishing the relationship between the student's knowledge state and the student's answer to problems of different target hierarchies to predict the hierarchy category in which the student is currently located.

To be specific, we first designed a KCH gate mechanism based on prior statistics of KCs correctness rate and frequency to first determine the hierarchy of the corresponding KCs. Then, we introduced the problem ID during the student's answering process to improve hierarchy embedding, and designed a network named FOTHLH. This network emphasizes the student's learning history by dynamically tracing the target hierarchy attention mechanism, allowing for a more refined division of target hierarchy. Following that, we designed a network KCAA for student knowledge application ability performance to capture the connection between the student's knowledge application ability and the target hierarchy in THKT. Then, we predicted the student's future performance by considering the relationship between the student's application ability and the target hierarchy at which the student is currently located. Finally, we conducted a large number of experiments to verify the effectiveness, adaptability, and interpretability of the proposed target hierarchy, and compared different layering methods for the target hierarchy, which demonstrated the rationality and necessity of layering.

We believe that the educational-based target hierarchy proposed in this paper should be applicable to other fields of smart education, such as cognitive diagnosis. However, we have not yet evaluated this method in other domains of smart education apart from KT tasks. In addition, the classification of the target hierarchy relies on prior statistics of the accuracy rate and frequency of KCs, and effective layering cannot be achieved for datasets without KCs. Furthermore, due to the current lack of gold-annotated data for the target hierarchy, further equivalence comparisons cannot be made. Although we successfully identified and classified the hierarchical structure of problems using our model, we have not yet been able to directly map these hierarchies to Bloom's Taxonomy. This limitation may restrict the breadth and depth of the application of our research findings. We leave this task to future work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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